

# Hausdorff dimension of recurrence sets

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# Outline

- 1 Background
- 2 Main results
- 3 Proofs

$(X, \mathcal{B}, T, \mu, d)$ : metric measure preserving system (m.m.p.s.)

- **Poincaré recurrence theorem** If  $(X, d)$  is a separable metric space, then  $\mu$ -a.e.  $x \in X$  is recurrent, that is

$$\liminf_{n \rightarrow \infty} d(T^n x, x) = 0.$$

- **Boshernitzan, 1993** Assume that for some  $\tau > 0$ ,  $\mathcal{H}^\tau(X)$  is  $\sigma$ -finite. Then for  $\mu$ -a.e.  $x \in X$ ,

$$\liminf_{n \rightarrow \infty} n^{\frac{1}{\tau}} d(T^n x, x) < \infty.$$

- Barreira & Saussol, 2001 If  $T: X \rightarrow X$  is a Borel measurable map on a measurable subset  $X \subset \mathbb{R}^m$ , and  $\mu$  is a  $T$ -invariant Borel probability measure on  $X$ , then for  $\mu$ -almost every  $x \in X$ , we have

$$\liminf_{n \rightarrow \infty} n^{1/\tau} d(T^n x, x) = 0$$

for any  $\tau > \underline{d}_\mu(x)$ , where

$$\underline{d}_\mu(x) = \liminf_{r \rightarrow 0} \frac{\log \mu(B(x, r))}{\log r}.$$

# Background

## Question

How large is the set of recurrent points when the rate of recurrence is replaced by a general function?

Let  $(X, \mathcal{B}, T, \mu, d)$  be a m.m.p.s. and  $r_n$  be some positive function on  $\mathbb{N}$ . Define the **recurrence set** as

$$E = \{x \in X : T^n x \in B(x, r_n) \text{ for infinitely many } n \geq 1\}.$$

- Tan & Wang(2011); Seuret & Wang(2015); Chang, Wu & Wu(2019); Baker & Farmer(2021); Hussain, Li, Simmons & Wang(2022); Kirsebom, Kunde & Persson(2023); He & Liao(2023); Kleinbock & Zheng(2023) ...

## Question

- $T$  is a linear mapping on  $\mathbb{T}^2$ .
- the rate  $r_n$  does not depend on the initial point.

How large is the set  $E$ , where

$$E = \{x \in \mathbb{T}^2 : T^n(x) \in B(x, r_n) \text{ for infinitely many } n \geq 1\}?$$

# Main results

## Theorem (Hu and Persson)

Let  $A$  be a  $2 \times 2$  integer matrix with  $|\det A| = 1$  and an eigenvalue  $|\lambda| > 1$ . Let  $T(x) = Ax \pmod{1}$ , and for  $n \geq 1$ ,  $r_n = e^{-\alpha n}$ ,  $\alpha > 0$ . Then

$$\dim_{\mathrm{H}} E = s_0,$$

where

$$s_0 = \min \left\{ \frac{2 \log |\lambda|}{\alpha + \log |\lambda|}, \frac{\log |\lambda|}{\alpha} \right\}.$$

Moreover, for  $\alpha > 0$ , we have  $E \in \mathcal{G}^{s_0}(\mathbb{T}^2)$ .

- **The large intersection property:** Given  $s \in (0, m]$ ,  $\mathcal{G}^s(\mathbb{R}^m)$  to be the class of all  $G_\delta$  sets  $F$  in  $\mathbb{R}^d$  such that the **Hausdorff dimension any set in  $\mathcal{G}^s(\mathbb{R}^m)$  is at least  $s$ , and closed under similarity transformations and countable intersections.**
- For a general non-increasing sequence of positive real numbers  $\{r_n\}_{n \geq 1}$ , let

$$\alpha = \liminf_{n \rightarrow \infty} \frac{-\log r_n}{n}.$$

Then the theorem also holds.

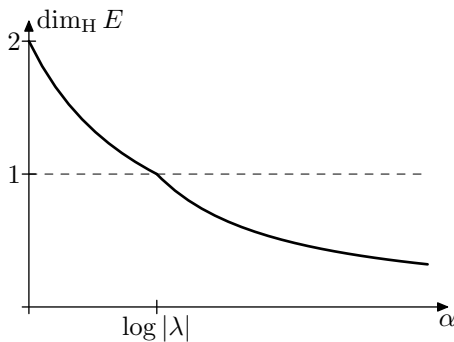


图: The graph of  $\dim_H E$  as a function of  $\alpha$ , with  $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ .

Let  $A$  and  $\{r_n\}_{n \geq 1}$  be as in the theorem.

$$E = \{x \in \mathbb{T}^2 : A^n x \pmod{1} \in B(x, r_n) \text{ i.o.}\}.$$

Denote

$$\begin{aligned}\mathcal{E}_n &:= \{x \in \mathbb{T}^2 : A^n x \pmod{1} \in B(x, r_n)\} \\ &= \{x \in \mathbb{T}^2 : (A^n - I)x \pmod{1} \in B(0, r_n)\}, \\ &= \{x \in \mathbb{T}^2 : x \in (A^n - I)^{-1} B(\mathbb{Z}, r_n)\},\end{aligned}$$

where  $I$  is the identity matrix. Then  $E = \limsup_{n \rightarrow \infty} \mathcal{E}_n$ .

- $\det A = 1$ ,  $\lambda > 1$ .

# Lower bound on $\dim_{\mathrm{H}} E$

- The  $s$ -dimensional **Riesz potential** of  $\mu$ :  $R_s \mu(x) = \int |x - y|^{-s} d\mu(y)$ .
- The  $s$ -dimensional **Riesz energy** of  $\mu$ :  
$$I_s(\mu) = \int R_s \mu d\mu = \iint |x - y|^{-s} d\mu(x) d\mu(y).$$

## Lemma (Persson, 2022)

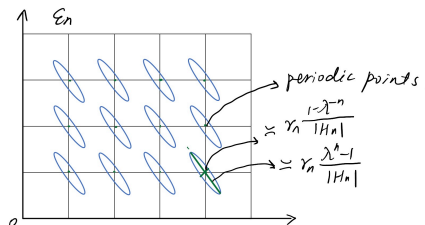
Let  $E_n$  be open sets in  $\mathbb{T}^m$  and let  $\mu_n$  be measures with  $\mu_n(\mathbb{T}^m \setminus E_n) = 0$ . If there is a constant  $C$  such that

$$C^{-1} \leq \liminf_{n \rightarrow \infty} \frac{\mu_n(B)}{\mathcal{L}(B)} \leq \limsup_{n \rightarrow \infty} \frac{\mu_n(B)}{\mathcal{L}(B)} \leq C$$

for any ball  $B$ , and  $I_s(\mu_n) < C$  for all  $n$ , then

$\limsup_{n \rightarrow \infty} E_n \in \mathcal{G}^s(\mathbb{T}^m)$ , and in particular we have  
 $\dim_{\mathrm{H}}(\limsup_{n \rightarrow \infty} E_n) \geq s$ .

# Structure of $\mathcal{E}_n$



The periodic points with period  $n$ :  $(A^n - I)x \pmod{1} = 0$ .

$\mathcal{E}_n$  consists of elliptical discs  $\{\mathcal{E}_n^i\}_i$  with semi-axes comparable to  $r_n \frac{\lambda^n - 1}{|H_n|} := \lambda_{n,1}$  and  $r_n \frac{1 - \lambda^{-n}}{|H_n|} := \lambda_{n,2}$ , whose centres are periodic points with period  $n$ , and  $H_n := \det(A^n - I)$ .

# Periodic Points

## Lemma

For every  $n \geq 1$ ,  $H_n = 2 - \lambda^n - \lambda^{-n}$ , and  $H_n$  is an integer.

## Lemma

The number of periodic points with period  $n$  is  $|H_n|$ , that is

$$\#\{x \in \mathbb{T}^2 : (A^n - I)x \pmod{1} = 0\} = |H_n|.$$

Proof:

$$\#\{x \in \mathbb{T}^2 : (A^n - I)x \pmod{1} = 0\} = \#\{x \in \mathbb{Z}^2 : x \in (A^n - I)\mathbb{T}^2\}.$$

Pick's theorem

# Periodic points when $n$ is odd

For  $n = 2k + 1$ , put

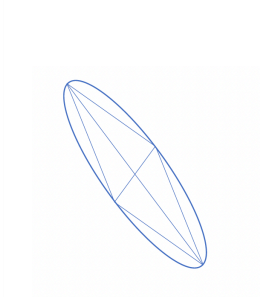
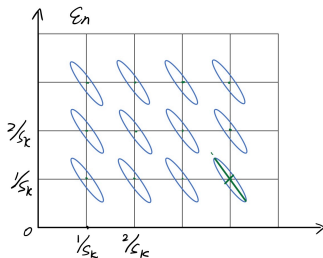
$$S_k = \lambda^k + \dots + \lambda + 1 + \lambda^{-1} + \dots + \lambda^{-k} = \frac{\lambda^{-k} - \lambda^{k+1}}{1 - \lambda}.$$

$S_k$  is an integer.

## Lemma

For  $n = 2k + 1$ ,  $k \geq 0$ , the points  $\left\{ \left( \frac{m}{S_k}, \frac{j}{S_k} \right) : m, j = 0, \dots, S_k - 1 \right\}$  are periodic points.

Proof: Prove that  $A'_{2k+1} := \frac{1}{S_k}(A^{2k+1} - I)$  is an integer matrix. This implies that if  $x = \frac{1}{S_k}[x_1 \ x_2]^t$  where  $x_1$  and  $x_2$  are integers, then  $(A^{2k+1} - I)x$  is an integer matrix, and hence  $T^{2k+1}(x) = x$ .



For  $i \geq 1$ ,  $\mathcal{E}_n^i$  contains a **parallelogram**  $E_n^i$  with vertices

$$x_{n,i} + \begin{bmatrix} 1 & \frac{\lambda-a}{b} \end{bmatrix}^t \lambda_{n,2},$$

$$x_{n,i} - \begin{bmatrix} 1 & \frac{\lambda-a}{b} \end{bmatrix}^t \lambda_{n,2},$$

$$x_{n,i} + \begin{bmatrix} 1 & \frac{\lambda^{-1}-a}{b} \end{bmatrix}^t \lambda_{n,1},$$

$$x_{n,i} - \begin{bmatrix} 1 & \frac{\lambda^{-1}-a}{b} \end{bmatrix}^t \lambda_{n,1},$$

the lengths of the diagonal lines of  $E_n^i$  are  $2\lambda_{n,1}$  and  $2\lambda_{n,2}$ .

# Lower bound on $\dim_H E$

## Lemma (Persson, 2022)

Let  $E_n$  be open sets in  $\mathbb{T}^m$  and let  $\mu_n$  be measures with  $\mu_n(\mathbb{T}^m \setminus E_n) = 0$ . If there is a constant  $C$  such that

$$C^{-1} \leq \liminf_{n \rightarrow \infty} \frac{\mu_n(B)}{\mathcal{L}(B)} \leq \limsup_{n \rightarrow \infty} \frac{\mu_n(B)}{\mathcal{L}(B)} \leq C$$

for any ball  $B$ , and  $I_s(\mu_n) < C$  for all  $n$ , then

$\limsup_{n \rightarrow \infty} E_n \in \mathcal{G}^s(\mathbb{T}^m)$ , and in particular we have

$\dim_H(\limsup_{n \rightarrow \infty} E_n) \geq s$ .

$n = 2k + 1$

- Case1:  $\alpha \geq \log \lambda / 2$ ;
- Case2:  $\alpha < \log \lambda / 2$ .

By **Liouville's theorem on diophantine approximation**, the distance between  $E_n^i$  and  $E_n^j$ ,  $i \neq j$ , satisfies

$$d(E_n^i, E_n^j) \begin{cases} \gtrsim \frac{1}{S_k^{|i_1-j_1|}} \gtrsim \lambda^{-n} > 0 & \text{if } i_1 \neq j_1, \\ \asymp S_k^{-1} & \text{if } i_1 = j_1, \end{cases}$$

where  $x_{n,i} = (\frac{i_1}{S_k}, \frac{i_2}{S_k})$ ,  $x_{n,j} = (\frac{j_1}{S_k}, \frac{j_2}{S_k})$ .

$\alpha$  is large,  $\alpha \geq \frac{1}{2} \log \lambda$

### Lemma

For  $\alpha \geq \frac{1}{2} \log \lambda$ , we have

$$\limsup_{k \rightarrow \infty} E_{2k+1} \in \mathcal{G}^{s_0}(\mathbb{T}^2),$$

where  $s_0 = \min \left\{ \frac{2 \log \lambda}{\alpha + \log \lambda}, \frac{\log \lambda}{\alpha} \right\}$ .

- $\mu_n = \frac{1}{\mathcal{L}(E_n)} \mathcal{L}|_{E_n}$ .
- $C^{-1} \leq \liminf_{k \rightarrow \infty} \frac{\mu_{2k+1}(B)}{\mathcal{L}(B)} \leq \limsup_{k \rightarrow \infty} \frac{\mu_{2k+1}(B)}{\mathcal{L}(B)} \leq C$ .
- $I_s(\mu_n)$  is finite for some  $s$ .

$\alpha$  is small,  $\alpha < \frac{1}{2} \log \lambda$

$\mathcal{L}^1$ : the Lebesgue measure restricted on  $\mathbb{T}$ .

Let  $L_{x_0} := \{ (x_0, y) : y \in \mathbb{T} \}$ ,  $x_0 \in \mathbb{T}$  and

$$F_n(x_0) = \{ \mathcal{E}_{n,i} \cap L_{x_0} : 1 \leq i \leq N_n, \lambda_{n,2}/4 \leq |E_n^i \cap L_{x_0}| \leq 2\lambda_{n,2} \}.$$

Assume that  $F_n(x_0) = \{ I_{n,i} : 1 \leq i \leq M_n \}$ . Let  $M_n = \#F_n(x_0)$  and

$$\mathcal{F}_n(x_0) = \bigcup_{i=1}^{M_n} I_{n,i}.$$

## Lemma

For  $0 < \alpha < \frac{1}{2} \log \lambda$ , given  $x_0 \in \mathbb{T}$ , we have

$$\limsup_{n \rightarrow \infty} \mathcal{F}_n(x_0) \in \mathcal{G}^{s_1}(L_{x_0}),$$

where  $s_1 = \frac{\log \lambda - \alpha}{\log \lambda + \alpha}$ . In particular, for any  $x_0 \in \mathbb{T}$ , we have

$$\dim \left( \limsup_{n \rightarrow \infty} \mathcal{F}_n(x_0) \right) \geq s_1.$$

## Lemma

Given any segment with  $L_I \subset L_{x_0}$ , of length  $\ell$ , for large enough  $n = 2k + 1$ , we have

$$\#\left\{ i : 1 \leq i \leq N_n, \frac{\lambda_{n,2}}{4} \leq |E_n^i \cap L_I| \leq 2\lambda_{n,2} \right\} \asymp \ell r_n \lambda^n.$$

In particular,  $\#M_n \asymp r_n \lambda^n$ .

For  $n = 2k + 1$ , let

$$\mu_n = \frac{1}{\mathcal{L}^1(\mathcal{F}_n)} \mathcal{L}^1|_{\mathcal{F}_n}.$$

- $C_3^{-1} \leq \liminf_{n \rightarrow \infty} \frac{\mu_n(B)}{\mathcal{L}^1(B)} \leq \limsup_{n \rightarrow \infty} \frac{\mu_n(B)}{\mathcal{L}^1(B)} \leq C_3.$
- $I_s(\mu_n) = \int \int |x - y|^{-s} d\mu_n d\mu_n$  is bounded for some  $s$ .

For  $p \geq 0$ , let

$$A_p = \left\{ j : I_{n,j} \cap \left\{ (x_0, y) : \frac{p+1}{S_k} > |y - x| \geq \frac{p}{S_k} \right\} \neq \emptyset \right\}.$$

Suppose that  $\mu$  is a finite measure on  $[0, 1]^2$  which can be disintegrated as

$$\mu(A) = \int \mu_x(A \cap \{x\} \times [0, 1]) d\nu(x) = \int \mu_x(A) d\nu(x),$$

for any  $A \in \mathcal{B}$ , and  $\text{spt} \mu_x \subset \{x\} \times [0, 1]$ . Assume moreover that  $\mu_x$  and  $\nu$  are probability measures.

### Lemma

With  $\mu$  as above, we have for  $x = (x_1, x_2)$  that

$$R_{s+t}\mu(x) \leq \int R_s\mu_{y_1}(x_2) |x_1 - y_1|^{-t} d\nu(y_1).$$

In particular, if the  $s$ -dimensional Riesz potentials of  $\mu_x$  are uniformly bounded and if  $\nu$  has a bounded  $t$ -dimensional potential, then

$$I_{s+t}(\mu) \leq \sup_x \|R_s\mu_x\|_\infty \|R_t\nu\|_\infty.$$

Let  $\tilde{\mu}_n$  be defined as

$$\tilde{\mu}_n(A) = \int \mu_{x,n}(A) \, dx,$$

for  $A \in \mathcal{B}$ , where  $\mu_{x,n} = \frac{1}{\mathcal{L}^1(F_n(x))} \mathcal{L}^1|_{F_n(x)}$ .

- $\{\mu_{x,n}\}_{x \in \mathbb{T}}$  is a disintegration of  $\tilde{\mu}_n$  over  $\mathcal{L}^1$ .
- For any  $\varepsilon > 0$ ,

$$I_{s_1+1-\varepsilon}(\tilde{\mu}_{2k+1}) < \infty.$$

# The upper bound on $\dim_{\text{H}} E$

## Lemma

Let  $A$  be a  $2 \times 2$  integer matrix with  $\det A = 1$  and an eigenvalue  $\lambda > 1$ . Let  $T(x) = Ax \pmod{1}$ , and for  $n \geq 1$ ,  $r_n = e^{-\alpha n}$ ,  $\alpha \geq 0$ . Then

$$\dim_{\text{H}} E \leq s_0,$$

where

$$s_0 = \min \left\{ \frac{2 \log \lambda}{\alpha + \log \lambda}, \frac{\log \lambda}{\alpha} \right\}.$$

# Thank you for listening!